GROUP MEETING · DATA STEWARDSHIP

FAIR Data Curation for Our Lab

Making our phenotyping data findable, usable, and ready for USDA and NSF data sharing

F

Findable
someone can locate it

A

Accessible
they can open it

I

Interoperable
their tools can read it

R

Reusable
they know how, and may

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The dataset that disappeared

A student finishes a UAV-LiDAR maize project. The paper says **"data available upon request."**

Two years later a reviewer asks for it. The student has graduated.

The shared link opens a folder like the one on the right.

```
final_v3_REAL/
  cloud_merged FINAL.las
  lidar_aug2_v3_REAL.laz
  lidar_aug2_v3_REAL (1).laz
  plot map FINAL.xlsx
  ~$plot map FINAL.xlsx
  Untitled.ipynb
  Untitled1.ipynb
  New folder/IMG_4521.HEIC
```

Questions nobody can answer now

- Which flight date is each cloud from?
- Which plot and genotype is which?
- Which coordinate system? Which software?
- Who is allowed to reuse it?

> The data still exists. It is just useless without the context that lived in one person's head.

Why now: our funders require it

EXCERPT · OSTP MEMORANDUM, AUG 25 2022 ("NELSON MEMO")

OSTP recommends that federal agencies, to the extent consistent with applicable law:

1. Update their public access policies as soon as possible, and no later than December 31st, 2025, to make publications and their supporting data resulting from federally funded research publicly accessible without an embargo on their free and public release;
2. Establish transparent procedures that ensure scientific and research integrity is maintained in public access policies; and,
3. Coordinate with OSTP to ensure equitable delivery of federally funded research results and data.

"Ensuring Free, Immediate, and Equitable Access to Federally Funded Research," White House OSTP (public domain). Retyped excerpt; highlight added.

€10.2 billion / year

Minimum yearly cost of non-FAIR research data to the EU economy (European Commission / PwC, 2018).

Aug 2022

OSTP "Nelson memo": papers **and supporting data**, no embargo

Dec 2024

Agencies publish new public-access policies

Dec 2025

Deadline for the new policies to take effect (USDA, NSF, NIH…)

2026

NeurIPS dataset track requires Croissant + Responsible-AI metadata

FAIR is a library catalog for data

Letter	In a library	In our lab
F Findable	Call number + catalog card	A **DOI** (permanent ID) + rich metadata in a searchable repository
A Accessible	Clear borrowing rules; the card stays even if the book is lost	Download over HTTPS; login allowed; metadata stays online
I Interoperable	Written in a language readers know	Open formats (LAS, ENVI, PLY, VCF, CSV) + shared trait terms
R Reusable	Edition, author, copyright page	License, provenance (how it was made), community standard (MIAPPE)

FAIR is not the same as open. Private industry genotypes can still be FAIR: the metadata is public, the DOI works, and the access rule is written down ("request through a data-use agreement").

The 15 principles: the yellow ones are our job

EXCERPT · THE FAIR GUIDING PRINCIPLES (WILKINSON ET AL. 2016)

TO BE FINDABLE

- **F1** (Meta)data are assigned a globally unique and persistent identifier
- **F2** Data are described with rich metadata (defined by R1 below)
- **F3** Metadata clearly and explicitly include the identifier of the data they describe
- **F4** (Meta)data are registered or indexed in a searchable resource

TO BE ACCESSIBLE

- **A1** (Meta)data are retrievable by their identifier using a standardised communications protocol
- **A1.1** The protocol is open, free, and universally implementable
- **A1.2** The protocol allows for an authentication and authorisation procedure, where necessary
- **A2** Metadata are accessible, even when the data are no longer available

TO BE INTEROPERABLE

- **I1** (Meta)data use a formal, accessible, shared, and broadly applicable language for knowledge representation
- **I2** (Meta)data use vocabularies that follow FAIR principles
- **I3** (Meta)data include qualified references to other (meta)data

TO BE REUSABLE

- **R1** (Meta)data are richly described with a plurality of accurate and relevant attributes
- **R1.1** (Meta)data are released with a clear and accessible data usage license
- **R1.2** (Meta)data are associated with detailed provenance
- **R1.3** (Meta)data meet domain-relevant community standards

Wilkinson et al., Sci Data 3:160018 (2016), CC BY 4.0; wording as listed at go-fair.org. Retyped; yellow = what our lab acts on, blue = why private data can still be FAIR.

Repository gives us

F1 DOI, F4 indexing, A1–A2 access and long-lived metadata

We must supply

F2/R1 rich metadata

I2 shared trait terms

R1.1 a license

R1.2 provenance

R1.3 MIAPPE / Croissant

Blue = private is OK

A1.2 login and approval are allowed

Written for machines, not only for people

EXCERPT · GO FAIR, "FAIR PRINCIPLES"

The principles emphasise machine-actionability (i.e., the capacity of computational systems to find, access, interoperate, and reuse data with none or minimal human intervention) because humans increasingly rely on computational support to deal with data as a result of the increase in volume, complexity, and creation speed of data.

go-fair.org/fair-principles (CC BY 4.0). Retyped excerpt; highlight added.

Search engines

Google Dataset Search and repository search read structured metadata, not your README prose.

Scripts and pipelines

A loader can pull every LiDAR file from session 03 only if names and metadata follow a rule.

AI agents

An agent can find, check, and load a Croissant-described dataset without emailing you.

How to score it: one weak letter sinks the whole dataset

Score each of the 15 sub-principles 0, 0.5, or 1.

Average

FAIR% = (100 / 15) × Σ s_k

Weakest link (teaching model)

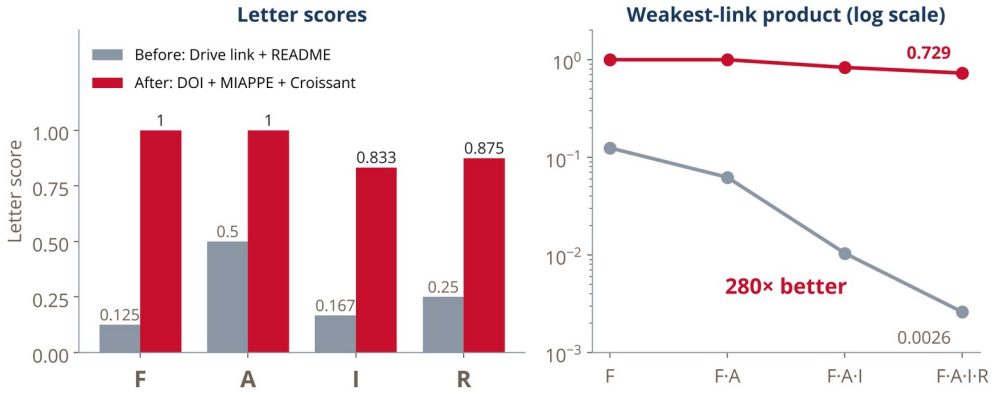
P(reuse) ≈ F × A × I × R

Reuse needs find AND open AND read AND trust, so the letters multiply.

Our LiDAR example

Mean: **26.7% → 93.3%** (3.5×)

Product: **0.0026 → 0.729** (280×)



Source: Worked example and scoring from our FAIR lesson; the product model is a teaching aid, not a published metric.

Rule 1: one numbered folder layout for every project

```
maize-uav-lidar-2025/
  README.md              <- start here
  00_docs/               notes, protocols
  01_metadata/           plot map, calibration,
                         manifest, MIAPPE, Croissant
  02_raw/                straight off the sensor
    01_uav_rgb/01_2025-07-14/
    04_lidar/02_2025-08-02/
  03_processed/          orthomosaics, clouds, 3DGS
  04_results/            trait tables, models
  05_code/               scripts, git commit
  99_review/             possible duplicates
```

Read the numbers in order

00 → 05 is the path a new student should follow: what it is, how to read it, the data, what we made from it, and the code.

Inside each folder

Data types are numbered in **pipeline order**: sensor data before the products made from it.

Raw is read-only

Never edit files in 02_raw. Everything else can be rebuilt from raw + code.

Rule 2: names carry the date, the session, and nothing else

- Dates in ISO 8601: **2025-07-14** in folders, **20250714** in names, so they sort in time order.
- Each field day gets one **session number** shared by every sensor: s01 = 2025-07-14 for drone, LiDAR, and photos.
- No spaces, and no **final**, **v3**, **REAL**, or **copy**. Versions live in git or the README.
- Keep sensor timestamp names (DJI_2025…); rename only meaningless ones (IMG_4521, Untitled1).

Before	After
flight july 14/DJI_20250714101502_0001_D.JPG	02_raw/01_uav_rgb/01_2025-07-14/DJI_20250714101502_0001_D.jpg
lidar/lidar_aug2_v3_REAL (1).laz	99_review/duplicates/…/lidar_aug2_v3.laz
misc/New folder/IMG_4521.HEIC	02_raw/06_photos/01_2025-07-14/s01_20250714_aea_photos_0001.heic
notes/plot map FINAL.xlsx	01_metadata/01_field_design/plot_map.xlsx

Rule 3: open formats, plus the context that must travel with them

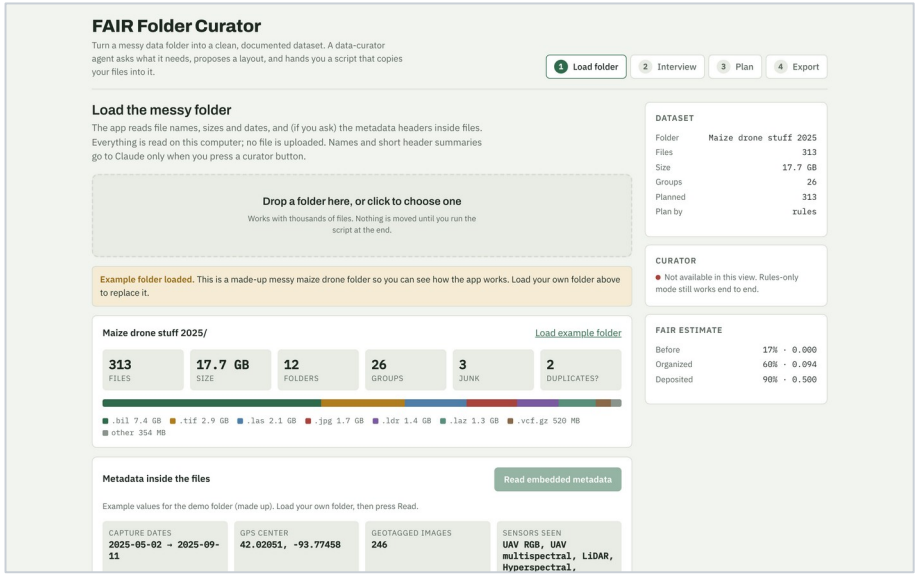
Data	Format	Must travel with it
UAV images	JPG / TIFF / DNG	EXIF + GPS kept, flight and RTK logs, calibration panel
LiDAR	LAS / LAZ	CRS (EPSG code) in the header, processing software and version
Hyperspectral	ENVI .hdr + .bil	Wavelength list, white and dark references
3D (SfM / NeRF / 3DGS)	PLY + transforms.json	Camera poses, training settings, code commit
Genotypes	VCF (.vcf.gz)	Reference genome version (e.g. B73 v5)
Traits	CSV / Parquet	Units and Crop Ontology trait IDs

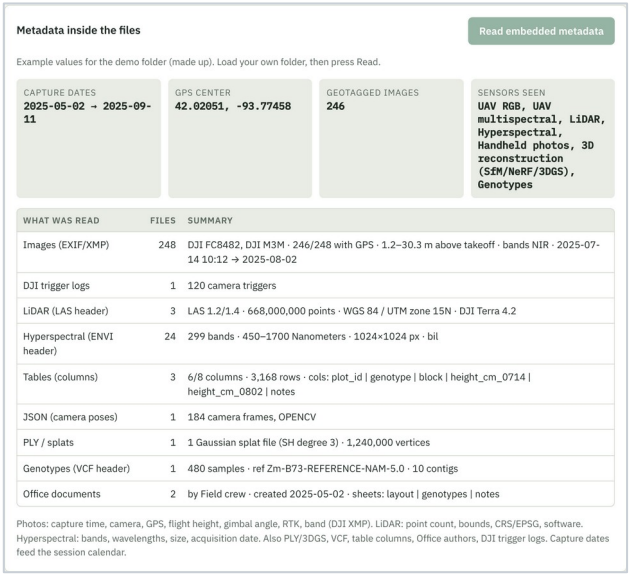
MIAPPE 1.1
Describes the experiment: site, design, plants, traits.

Croissant
Describes an ML-ready dataset in JSON-LD.

Crop Ontology
Shared IDs so "plant height" means one thing.

Demo: FAIR Folder Curator reads the messy folder





Drop a folder in. It reads names, sizes, dates, and file headers (GPS, capture time, LiDAR CRS, band lists) **on your computer; nothing is uploaded**. A data-curator agent then asks 3–7 questions about what it cannot see.

Demo: it plans the layout and hands you a safe script

New folder tree — 313 files · 3 skipped · 2 to review

```
maize-uav-lidar-2025/
  ▼ 00_docs/ 1 files · 380 KB
    01_notes/ 1 files · 380 KB
  ▼ 01_metadata/ 2 files · 7.0 MB
    01_field_design/ 1 files · 56.0 KB
    02_calibration/ 1 files · 6.9 MB
  ▼ 02_raw/ 299 files · 11.8 GB
    ▶ 01_uav_rgb/ 216 files · 1.7 GB
    ▶ 02_uav_multispectral/ 24 files · 62.4 MB
    ▶ 03_flight_logs/ 2 files · 358 KB
    ▶ 04_lidar/ 2 files · 2.0 GB
    ▶ 05_hyperspectral/ 48 files · 7.4 GB
    ▶ 06_photos/ 5 files · 12.0 MB
    07_weather/ 1 files · 220 KB
    08_genotypes/ 1 files · 520 MB
  ▼ 03_processed/ 5 files · 5.2 GB
    01_orthomosaics/ 2 files · 2.8 GB
    02_lidar_processed/ 1 files · 2.1 GB
    03_3d_reconstruction/ 1 files · 120 KB
    04_point_clouds/ 1 files · 340 MB
```

Session calendar — Each field day gets one number, shared by every sensor

SESSION	DATE	DATA COLLECTED
01	2025-07-14	uav_rgb, flight_logs, lidar, photos
02	2025-08-02	uav_rgb, uav_multispectral, lidar
03	2025-09-10	hyperspectral
04	2025-09-11	hyperspectral

```
# 1. preview: changes nothing
python organize.py "messy" "clean"
# 2. copy into the new layout
python organize.py "messy" "clean" --apply
# 3. changed your mind?
python organize.py --undo
clean/01_metadata/manifest.csv
```

- Copies by default; your originals stay put.
- manifest.csv maps every old path to its new one.
- Also writes README, MIAPPE, Croissant, FAIR report.

Last mile: license, deposit, check, cite

License
CC BY 4.0 for data, MIT for code. No license means nobody may legally reuse it.

Deposit (this gives the DOI)
ISU DataShare · Zenodo · CyVerse Data Commons · Hugging Face (ML sets) · NCBI SRA (reads)

Check
Run the DOI through **F-UJI**, a free automated FAIR checker; fix what fails.

Cite
Put the DOI in the paper's data-availability statement. Papers linking data in a repository saw up to **25% more citations** (an association, not proof).

FAIR score — 15 sub-principles scored 0, 0.5 or 1

SUB-PRINCIPLE	BEFORE	AFTER ORGANIZING	AFTER DEPOSIT
F1 Persistent ID (DOI)	0	0	1
F2 Rich metadata	0	1	1
F3 Metadata names the data ID	0	0.5	1
F4 Indexed / searchable	0	0	1
A1 Retrievable by ID	0.5	0.5	1
A1.1 Open protocol	1	1	1
A1.2 Access rules allowed	0.5	0.5	1
A2 Metadata outlives data	0	0	1
I1 Formal metadata language	0.5	1	1
I2 Shared vocabularies	0	0.5	0.5
I3 Links to related data	0	0	0
R1 Many relevant attributes	0	1	1
R1.1 Clear license	0	1	1
R1.2 Provenance	0	1	1
R1.3 Community standard	0	1	1
Mean of 15	17%	60%	90%
F·A·I·R (weakest link)	0.000	0.094	0.500

"After deposit" assumes you upload the organized folder to a repository that gives a DOI (ISU DataShare, Zenodo, CyVerse). It is a projection, not a measurement.

Source: Colavizza et al., PLOS ONE 2020; Devaraju & Huber, Patterns 2021 (F-UJI).

Your job before next group meeting

- **Pick** one dataset you own.
- **Run** it through FAIR Folder Curator and answer the interview.
- **Preview**, then **--apply** organize.py (it copies; originals stay).
- **Fill** the README blanks: processing steps, license, trait IDs.
- **Bring** your before and after FAIR score.

Everything is in one repo

github.com/znjubery/fair-data-curation

This talk (PPTX, Beamer, PDF), the Folder Curator app, the FAIR lesson, and the excerpt images.

FAIR in one line: a stranger, or a script, can find, open, understand, and legally reuse your data without emailing you.